Simon Degala Campos

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SUMMARY

Senior Software Engineer | Backend & Python Expert with 6+ years of specialization in designing and building scalable, high-performance backend systems. Skilled AI/ML Engineer with a strong track record in developing robust AI agents and intelligent solutions. Proficient in creating efficient, scalable architectures that drive innovation and support complex AI workflows. Adept at leveraging cutting-edge technologies to deliver impactful, reliable backend services for demanding applications.

SKILLS

AL & ML

PyTorch, TensorFlow, Keras, JAX, Hugging Face, LangChain, RAG, LlamaIndex, Numpy, Pandas, Polars, Scikit-learn, Jupyter, OpenCV, XGBoost, MLflow, Streamlit

Frontend

React, Next.js, JavaScript, TypeScript, HTML/CSS, Tailwind CSS, Zustand, Vite 6, Yarn, Figma

Backend

Python, Django, FastAPI, Flask, Node.js, Express.js, Go, Rust, RESTful API, GraphQL, WebSockets, Microservices, MySQL, PostgreSQL, MongoDB, Redis, Supabase

Mobile

React Native, Expo, Flutter, Dart, Kotlin, Swift, Android Studio, Xcode, Firebase

DevOps

Git, Github, Github Actions, AWS, Google Cloud, Kubernetes, Docker, Terraform, Kafka, Grafana, RabbitMQ, Gitlab

Trading

Timeseries Forecasting (LSTM, TimeXer, TimeMixer, TimeMixer++, TabPFN-v2, TIME-MOE), Sentiment Analysis, Pattern Detection, Algorithm Trading, Backtesting, Optimization, Cryptocurrency, Forex Trading, Stock Market, Trading Automation (TradingView, IBKR, MetaTrader 4, MetaTrader 5, NinjaTrader 8, cTrader, Binance, Bybit, Bitget, Blofin, MEXC, TradeStation)

Other

Claude, Cursor, Postman, Selenium, Cypress, OpenAI API,

EXPERIENCE

Founder & CTO

PXI Labs LLC

02/2026 - Present Part-Time

PXI Studio is a social and creative workspace platform: users get a ranked media feed, events and ticketing, albums with threaded discussion, chat, scrapbooks, a vault, notifications, and vendor/event-staff flows - delivered as a native mobile app, a marketing and dashboard web app, and a dedicated API tier.

- Led engineering end to end as Founder / CTO / Lead Software Engineer: owned system design across three codebases (Expo/React Native mobile, Next.js web, Node/Express API), data model split (relational + document), auth strategy, media pipeline, payments, and deployment story (e.g. API on EC2 with Docker/CI per repo docs; web on Netlify with Edge middleware).
- Designed and shipped a TypeScript/Express backend with REST APIs for auth, feed, social graph, events, albums, chat, media, scrapbooks, vault, tickets, vendors, safety, and webhooks; PostgreSQL (Prisma) + MongoDB (Mongoose) for the right data on the right store; Socket.io for real-time behavior where the product needs it.
- Implemented security-sensitive auth: PASETO v4 token issuance and verification, integration with Next.js Edge middleware and public-key verification for dashboard RBAC without extra round-trips; social login (Google/Apple) and session patterns wired through mobile and web clients.
- Built feed and engagement mechanics including Wilson score-based ranking, reactions, and notification flows so discovery and quality signals are first-class in the API and clients.
- Owned media and storage architecture using AWS S3 (upload SDKs, presigned flows) so images/video and related

assets scale; paired with mobile capture/edit UX and backend media routes.

- Integrated monetization and operations: Stripe (including Connect-style flows and a unified webhook surface for platform + connected accounts), plus transactional email (Resend) and SMS/voice (Twilio) where the product requires it.
- Shipped the mobile app on Expo / React Native (Expo Router, TypeScript, NativeWind): tabs for core surfaces (e.g. wall, studio, camera, passport), secure token storage, socket.io-client, Stripe React Native, camera stack (react-native-vision-camera), and advanced imaging (React Native Skia shaders/LUT-style processing for the "studio" camera pipeline).
- Delivered the web property on Next.js (App Router) with public marketing/event pages and a middleware-protected dashboard (passport, vendor upgrade, account); Ant Design, charts (Recharts), and Stripe.js for web payments; aligned with backend "soft gate" vs verified-user rules documented in the API guide.
- Established how the team ships: API and DB documentation, dev workflow (branching/PR policy), and CI/CD notes so contributors can run, migrate, and deploy consistently—reducing bus factor as the org grows.

AI/ML Engineer

Freelancer

06/2024 - 02/2026 Remote

AI/ML & Full-Stack Engineer with 10+ client projects spanning AI-powered web/mobile apps, trading systems, and automation tools. Combines software engineering expertise with practical AI/ML implementation skills

- Built end-to-end web applications with React/Next.js on the frontend and Node.js or Python backends, exposing REST/GraphQL APIs and integrating real-time features where needed
- Developed AI-driven features such as chatbots, recommendation flows, and predictive dashboards by wiring custom or third-party models into backend services and frontend UX
- Created cross-platform mobile apps using Flutter and/or React Native, integrating backend APIs and, in some cases, lightweight on-device inference or remote model calls for AI functionality
- Designed and implemented several algorithmic trading systems for crypto/forex, including data ingestion, strategy backtesting, live execution, and risk controls, using Python and Go on the backend
- Worked with common ML libraries and services (TensorFlow/PyTorch, scikit-learn, or hosted APIs) to build or integrate models for forecasting, classification, and basic NLP tasks, then deployed them behind secure APIs
- Containerized backend services with Docker and used CI/CD workflows to deploy to AWS/GCP or other hosting platforms, helping clients ship updates more frequently with fewer manual steps
- Collaborated closely with non-technical clients to clarify requirements, propose technical approaches, estimate work, and provide regular progress updates, ensuring projects were delivered on time and within scope

Senior Backend Developer

Plentina VN

08/2022 - 05/2024 Paranaque, Philippines

Plentina is a fintech startup providing Buy Now, Pay Later (BNPL) credit solutions in emerging markets, primarily in the Philippines. The company helps consumers build credit history and make cashless payments through a mobile app and partnerships with merchants

- Led backend development for core credit and loan services, including onboarding, credit assessment, repayment schedules, and transaction processing.
- Designed scalable backend systems to handle higher user and transaction volume.
- Built and maintained high-throughput REST APIs for the mobile app, internal tools, and partner integrations.
- Integrated risk scoring, eligibility checks, and machine learning outputs into backend decision flows.
- Designed secure data flows for sensitive financial information, including encryption, access control, and audit logging.
- Tuned MySQL/PostgreSQL queries and indexes to improve performance for high-volume BNPL transactions.
- Managed integrations with payment gateways, e-wallets, and merchant systems, including failure handling and reconciliation.
- Owned incident response, monitoring, and root-cause analysis for production backend issues.
- Worked with product, data, and mobile teams to estimate work, plan releases, and improve system architecture.
- Helped refactor the backend from a more monolithic setup into a cleaner, modular service structure.

Junior Backend Developer

Plentina VN

04/2021 - 07/2022 Paranaque, Philippines

Plentina is a fintech startup providing Buy Now, Pay Later (BNPL) credit solutions in emerging markets, primarily in the Philippines. The company helps consumers build credit history and make cashless payments through a mobile app and partnerships with merchants

- Built and maintained small backend features for credit and loan workflows, such as user onboarding, repayment updates, and transaction records.
- Developed and tested REST API endpoints for the mobile app and internal tools.
- Fixed bugs, handled minor production issues, and supported day-to-day backend maintenance.
- Worked with Node.js/TypeScript, PHP/Laravel, REST APIs, Kafka, Kubernetes, and Docker under senior guidance.
- Wrote unit tests, updated documentation, and followed coding standards set by the backend team.
- Assisted with staging deployments and basic CI/CD tasks.

Backend Developer • Intern

Converge ICT

10/2019 - 04/2021 Angeles, Philippines

Converge ICT Solutions Inc. is a premier fiber optic internet and digital services provider in the Philippines, known for its extensive fiber network and commitment to reliable, high-speed internet connectivity for homes and businesses

- Developed and maintained backend features for internet service provisioning and billing platforms using Node.js, PHP, and related frameworks, helping support day-to-day operations and improve the overall customer experience
- Worked on MySQL and PostgreSQL databases by designing and refining schemas, writing optimized queries, and handling large sets of customer and service data to reduce query times and improve application responsiveness.
- Built and integrated secure RESTful APIs with third-party monitoring, provisioning, and billing systems, focusing on reliable data exchange, proper error handling, and consistent API documentation
- Investigated and resolved backend performance issues, such as slow endpoints and inefficient queries, and helped address security concerns by applying patches, tightening access controls, and following internal security guidelines
- Collaborated with senior engineers to design and implement modular features, participated in sprint planning and stand-ups, and followed agile practices to deliver updates in small, manageable increments
- Contributed to the CI/CD pipeline by writing and maintaining build scripts, adding basic automated tests, and assisting with deployments to staging and production environments
- Performed code reviews for peer changes, followed established coding standards, and helped refactor legacy code to make it easier to maintain and extend
- Assisted in upgrading and migrating parts of the legacy backend stack to more scalable and maintainable solutions, helping the team support new services, higher traffic, and additional business requirements

EDUCATION

Bachelor's Degree in Computer Science

University of Rizal System (URS)

2016 - 2019 Tanay, Philippines

- Completed coursework in software development, data structures, algorithms, databases, and programming languages
- Participated in projects involving web development, mobile apps, and software design
- Gained practical experience in coding, debugging, and collaborative development environments

PROJECTS

PXI Studio - Social Event Platform (Full-Stack: Mobile, Web & API)

https://www.pxispace.com/

https://apps.apple.com/app/pxi/id6751762197

A multi-client platform for events, media, and community: ranked social feed, albums and threaded discussion, real-time chat and notifications, ticketing and vendor/staff flows, scrapbooks, vault, and account verification—delivered as a native mobile app, a Next.js marketing and dashboard site, and a dedicated Node API.

Technical Features

- Led end-to-end architecture across three repositories (Expo/React Native, Next.js, Express/TypeScript) with a clear split between product surfaces and a shared REST + real-time backend.
- Built a TypeScript Expres API with PostgreSQL (Prisma) and MongoDB (Mongoose) so relational identity/settings and high-churn social/document data each sit on an appropriate store; documented APIs, schema, and deployment for the team
- Implemented PASETO v4 token auth and wired Next.js Edge middleware to verify tokens and enforce dashboard RBAC without extra round-trips; integrated Google/Apple sign-in and mobile secure token storage
- Shipped a Wilson score-ranked media feed with reactions, notifications, and social graph primitives (follows, friend requests, safety/reporting patterns as implemented in routes)
- Added Socket.io for real-time behavior alongside REST, plus AWS S3-backed media upload and delivery patterns for images and user-generated content
- Integrated Stripe (including Connect-oriented flows and a unified webhook surface), plus Resend and Twilio for operational messaging where the product requires it
- Developed the Expo/React Native app with Expo Router, TypeScript, and NativeWind: camera capture, advanced on-device imaging (React Native Skia / vision-camera stack), albums and thread UI, checkout hooks, and socket client integration
- Delivered the Next.js (App Router) web app with public pages and middleware-protected dashboards (passport, vendor upgrade, account), Ant Design UI, charts (Recharts), and Stripe.js for web payments
- Established CI/CD (e.g. API on EC2 with Docker/GitHub Actions per project docs; web on Netlify), environment conventions, and contributor workflow so releases stay repeatable

Technology Stack

TypeScript, Node.js, Express, Prisma, PostgreSQL, MongoDB, Mongoose, PASETO, Socket.io, AWS S3, Stripe, Next.js, React, React Native, Expo, NativeWind, Ant Design, Docker, GitHub Actions, Netlify (Edge middleware)

FxUtopia - Forex & Crypto Trading Hub API (High-Performance Python Backend)

https://www.fxutopia.com/

A comprehensive FastAPI backend for a Forex & Crypto trading community platform featuring authentication, community forums, educational content, marketplace, reviews, signals, and economic calendar.

Technical Features

- Architected a modular FastAPI backend for a Forex & Crypto trading platform using Clean Architecture with repository/service layers to keep business logic isolated and easy to test
- Implemented secure auth flows with JWT access/refresh tokens, Google/GitHub OAuth, email verification, password reset, 2FA, and role-based access control
- Built community features including posts, threaded comments, reactions, user badges, and WebSocket chat, with pagination, caching, and full-text search for scalable read patterns
- Delivered education modules for courses, articles, and videos with progress tracking, completion certificates, and a searchable knowledge base
- Developed a marketplace for digital trading tools (EAs, bots, indicators) covering product listing APIs, secure file upload/delivery, and integrated ratings and reviews, designed to plug into a payment gateway
- Implemented broker/platform review flows with verified reviews, detailed rating dimensions, and “helpful” votes to surface quality content
- Exposed trading signal and provider APIs with subscription management, performance stats, leaderboards, and WebSocket endpoints for near real-time distribution
- Added an economic calendar API for macro events with impact levels, country/currency filters, and live update support
- Inter-Service Communication: gRPC protocol implementation
- Used Redis for caching and Celery workers/beat for background and scheduled tasks (emails, analytics, file processing) to keep the main API responsive
- Designed the relational data model in PostgreSQL and managed migrations with SQLAlchemy + Alembic, including indexes and constraints aligned with query access patterns
- Introduced structured logging, metrics, centralized error handling, and rate limiting plus security middleware (CORS configuration, strict validation) for production readiness
- Containerized services with Docker and Docker Compose, wiring up API, PostgreSQL, Redis, and Celery into a reproducible local and deployment setup

Technology Stack

Python, Django, FastAPI, PostgreSQL, Redis, SQLAlchemy, Alembic, Celery, JWT, Pydantic, Docker

AskYourPDF - Document Intelligence & RAG Platform (Python / FastAPI Backend + Multi-Client Product)

- https://askyourpdf.com/
- https://askyourpdf.com/developers
- https://apps.apple.com/app/askyourpdf-ai-chat-with-pdf/id6462732878
- https://chromewebstore.google.com/detail/askyourpdf/gaieenaffioioljpcocdkpphlifbhaig

A document-AI platform used at scale for chatting with PDFs and other files, summarization, knowledge-base Q&A, smart autofill, and embeddable website agents—delivered across web, iOS/iPad, Chrome extension, and a public developer API, with multi-model LLM support (GPT, Claude, Gemini), agent mode, source citation, and streaming responses.

Technical Features

- Architected a Python/FastAPI backend for document intelligence using a RAG pipeline: upload → parse → chunk → embed → retrieve → generate, keeping answers grounded in uploaded PDFs and knowledge bases
- Built document ingestion and processing for PDF/text workflows, including chunking, metadata handling, and vector indexing for semantic search over user documents and multi-document knowledge bases
- Exposed REST APIs for document upload, single-document chat, multi-document knowledge-base chat, summarization, and streaming (SSE) responses with API-key authentication and usage metering
- Implemented an agent orchestration layer for higher-quality multi-step RAG (retrieve → reason → cite) instead of single-pass retrieval-only answers
- Added a citation service so responses include page references and source links, improving trust and verifiability for research and enterprise use cases
- Routed requests across multiple LLM providers (OpenAI GPT, Anthropic Claude, Google Gemini) with per-request model selection and multi-model fallback patterns
- Delivered developer-facing API documentation, rate limits, and tiered access for third-party integrations and product clients
- Powered first-party clients from one shared backend: Next.js web app, React Native iOS/iPad app, Chrome extension (Manifest V3), embeddable website agent widget, Zotero plugin, and ChatGPT plugin
- Built web product features: document library management, multi-file knowledge-base chat, PDF autofill from paired source documents, summaries, and account/subscription flows
- Shipped cross-platform mobile app (React Native / TypeScript) with document upload, streaming chat, multi-document

sessions, literature-review/summarization tools, and unified web/mobile subscriptions
- Published Chrome extension (20,000+ users, 4.5★) for in-browser PDF chat, summaries, and Q&A without leaving the workflow
- Used PostgreSQL for users, document metadata, and API keys; Redis for caching and task queues; AWS S3 for raw document storage; Celery workers for async ingestion and background jobs
- Containerized services with Docker and CI/CD for reproducible local dev and deployment
- Applied GDPR-oriented data handling, document isolation per account, encrypted transport, and API-key security for developer access

Technology Stack

Python, FastAPI, RAG, Vector Embeddings, OpenAI API, Claude, Gemini, LangChain, PostgreSQL, Redis, AWS S3, Celery, Docker, REST API, SSE/Streaming, React, Next.js, TypeScript, React Native, Chrome Extension (Manifest V3), GDPR

Dr.Oracle - Medical AI Platform (Python / FastAPI Backend + React Native Mobile)

- https://droracle.ai/
- https://apps.apple.com/us/app/dr-oracle-ai-medical-assistant/id6473809057
- https://play.google.com/store/apps/details?id=ai.thedeep.droracleandroid

Physician-owned medical AI platform (TheDeep LLC) for students, residents, and clinicians—delivering evidence-based clinical Q&A with in-text citations from PubMed, international guidelines, FDA drug labels, and case reports. Shipped as native iOS/iPad and Android apps backed by a medical RAG API, with Research Mode, differential diagnosis support, guideline-directed management, and streaming cited responses. Adopted by hospital systems, residency programs, and universities.Plante AI is a Flutter app that uses AI to analyze meals from photos, providing detailed nutritional information and calorie estimates.

Technical Features

- Architected a Python/FastAPI medical RAG backend that ingests and indexes PubMed literature, clinical guidelines, FDA drug labels, and case reports into a vector retrieval pipeline so answers stay grounded in citable sources
- Built clinical chat APIs for real-time medical Q&A with in-text citation linking, streaming (SSE) responses, and conversation history synced to mobile clients
- Implemented Research Mode for literature synthesis—evaluating study strengths, weaknesses, and evidence quality across multiple clinical papers
- Delivered differential diagnosis and guideline-directed treatment workflow endpoints to support complex case reasoning on mobile
- Integrated PubMed / NCBI E-utilities for live biomedical literature retrieval and cross-referencing against indexed medical knowledge
- Added a citation engine plus fact-check and self-correction layers to improve accuracy and traceability of generated clinical content
- Routed queries through a fine-tuned medical LLM with multi-step RAG orchestration for general medicine and research workflows
- Established a physician advisory review queue and weekly model fine-tuning pipeline for ongoing accuracy improvements
- Shipped React Native mobile apps for iPhone, iPad, and Android: document-style clinical chat, Research Mode, exportable answers, haptic feedback, and subscription-aware UX
- Wired mobile clients to shared REST APIs with secure session auth, push-ready architecture, and unified conversation/subscription state
- Integrated mobile monetization via Apple In-App Purchases and Google Play Billing, with backend subscription and rate-limit enforcement
- Used PostgreSQL for users, subscriptions, and chat history; Redis for caching and task queues; AWS S3 for document/assets storage; Celery workers for async ingestion and background jobs
- Containerized services with Docker and CI/CD for reproducible deployment; encrypted transport and per-user session isolation for sensitive medical workflows

Technology Stack

Python, FastAPI, RAG, LangChain, PubMed, NCBI E-utilities, Vector Embeddings, LLM, Medical NLP, PostgreSQL, Redis, AWS S3, AWS CloudFront, Celery, Docker, REST API, SSE/Streaming, React Native, TypeScript, iOS, iPadOS, Android, In-App Purchases, Push Notifications